

Incomplete reversibility of fixation-induced changes in postmortem human brain WM MRI following rehydration



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Background and Objective

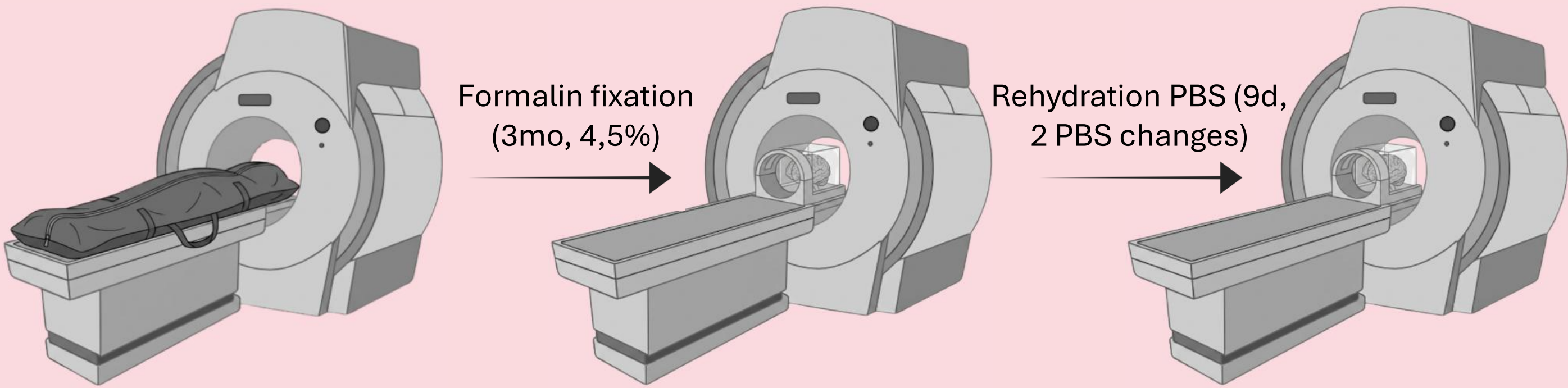
- Validation of MRI biomarkers: based on formalin-fixed ex situ imaging and histology
- Fixation: alters relaxation properties and tissue anisotropy¹⁻⁴
- Rehydration: suspected to partially reverse fixation effects → irreversible protein cross-linking suggests incomplete reversibility⁴⁻⁵

Objective: Assess effects of formalin fixation and rehydration on MRI derived measures in human brain WM

Methods

Subjects & MRI:

- Two human brains scanned on Siemens Prisma 3T



MRI Protocol:

- MP2RAGE: structural imaging and tissue segmentation
- Multi-echo GRE (12 echoes): R_2^* mapping
- Multi-shell DTI ($b = 0, 1000, 2000, 3000 \text{ s/mm}^2$)

Image processing:

- HD-BET⁶ - brain extraction, FSL FAST⁷⁻¹⁰ - segmentation, FSL eddy_correct⁷⁻⁹, FSL DTIFIT⁷⁻⁹ - MD, FA, AD, RD

Analysis:

- Fiber angles derived from DTI V1, binned $18 \times 5^\circ$ ($0^\circ - 90^\circ$)
- R_2^* calculated voxel-wise, data normalised and registered
- R_2^* anisotropy index $AI = \frac{(R_2^{*max} - R_2^{*min})}{(R_2^{*max} + R_2^{*min})}$

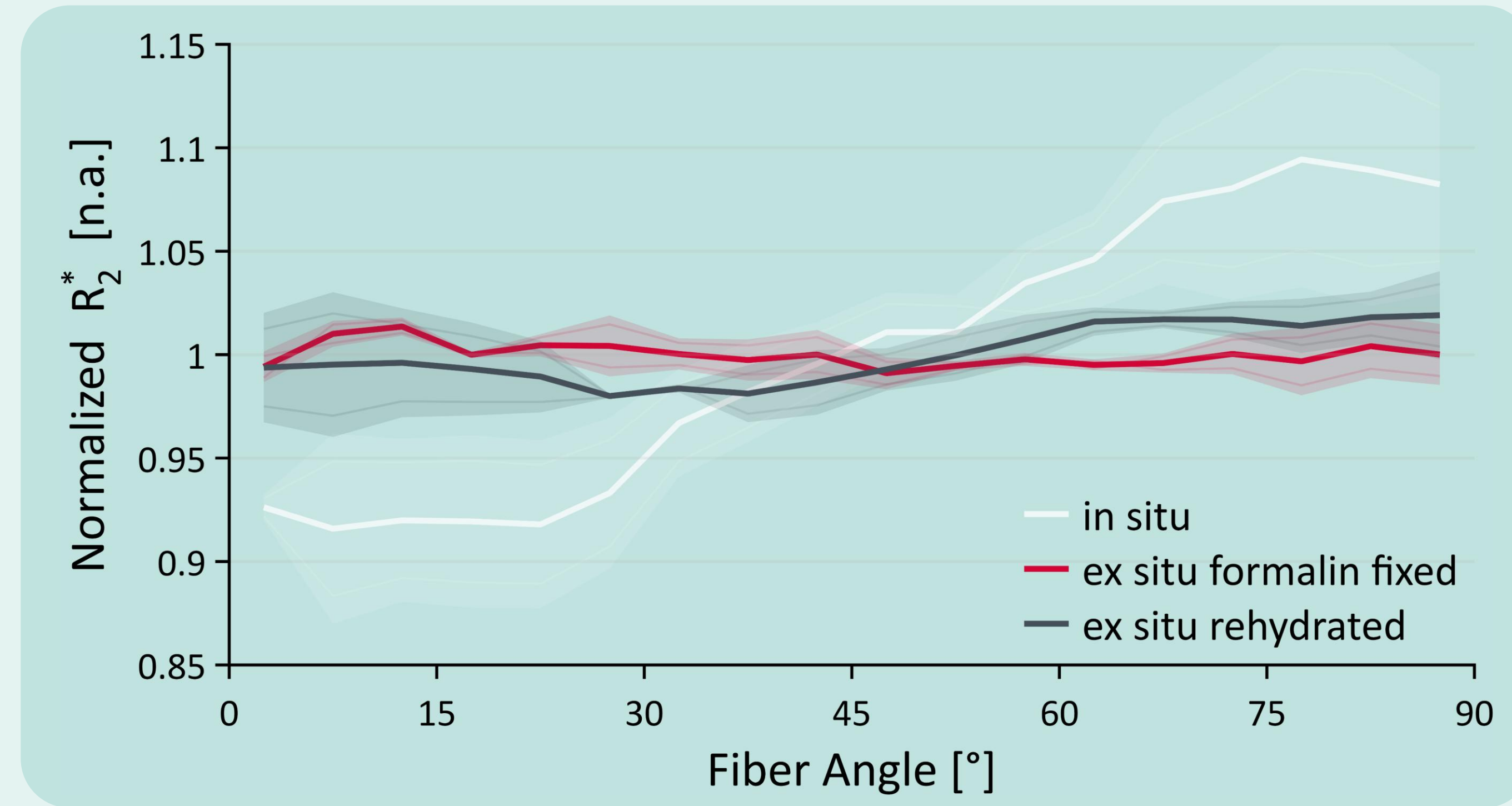
Results and Discussion

- Formalin fixation alters WM microstructure
- Fixation: FA ↓, MD ↓, AD ↓, RD ↓, AI ↓, R_2^* ↑
- Rehydration: FA ↓, MD ↑, AD ↑, RD ↑, AI ↑, R_2^* ↓
- Angular dependency: in situ > rehydrated > fixed

Table 1: Diffusion parameters FA, MD, AD and RD, as well as R_2^* and anisotropy index across the three tissue conditions.

	In situ	PM ex situ formalin-fixed	PM ex situ rehydrated
FA	0.47 ± 0.02	0.37 ± 0.05	0.30 ± 0.00
MD [$10^{-5} \text{ mm}^2/\text{s}$]	16.22 ± 6.48	13.03 ± 1.34	15.26 ± 2.84
AD [$10^{-5} \text{ mm}^2/\text{s}$]	24.80 ± 9.82	17.54 ± 2.23	19.87 ± 4.01
RD [$10^{-5} \text{ mm}^2/\text{s}$]	12.10 ± 4.71	11.01 ± 1.09	12.95 ± 2.25
R_2^* [s^{-1}]	27.79 ± 4.48	32.39 ± 0.38	28.40 ± 0.40
AI [%]	9.3 ± 3.3	1.5 ± 0.0	2.8 ± 0.4

Figure 1: Normalized R_2^* values across the three tissue conditions over the calculated tissue fiber angle.



Formalin fixation:

- Diffusivity, FA and R_2^* orientation dependency: ↓

Rehydration:

- partial recovery of MD, AD, AU and R_2^*
- FA remained reduced, RD exceeded in situ → Tissue properties not fully restored

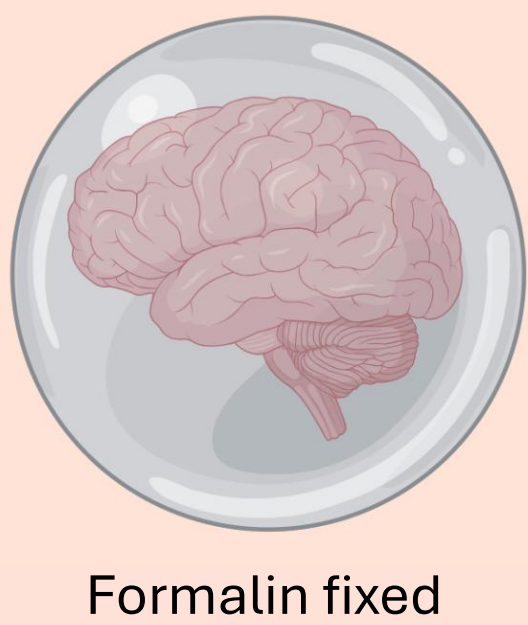
Key Message: Fixation effects are partly reversible, but not fully recoverable through rehydration

Conclusion

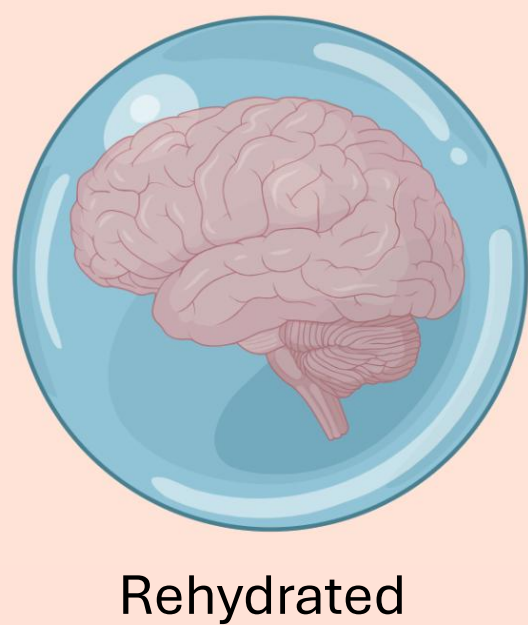
- Fixation and rehydration:
 - substantially alter MRI-derived measures
 - heterogeneous & partial reversibility



In situ



Formalin fixed

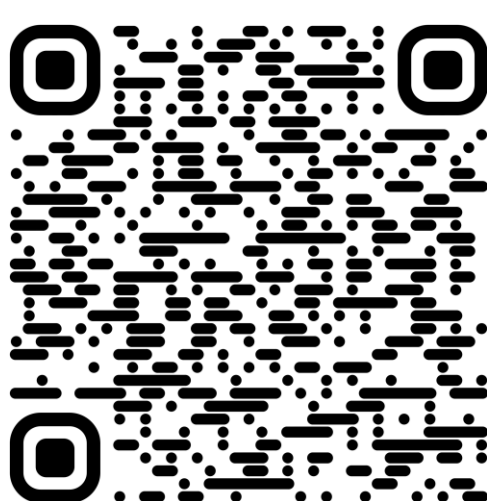


Rehydrated

Comparison of in vivo MRI to ex situ measurements for biomarker validation is limited

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